

UCI: Division of Continuing Education
Accelerated Certificate Program (ACP): Data Science
Introduction to Python Programming
 I&C SCI_X426.58:
Homework#5

Date Given: April 11, 2023

Due Date: April 12, 2023

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There are 6 problems in this homework assignment. Write a program in Python to solve these problems.

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The following problems have been taken from a book on MATLAB. Replace the word MATLAB with Python in these problems.

5. Type this matrix in MATLAB and use MATLAB to carry out the following instructions.

$$A = \begin{bmatrix} 3 & 7 & -4 & 12 \\ -5 & 9 & 10 & 2 \\ 6 & 13 & 8 & 11 \\ 15 & 5 & 4 & 1 \end{bmatrix}$$

- a. Create a vector **v** consisting of the elements in the second column of **A**.
 - b. Create a vector **w** consisting of the elements in the second row of **A**.
6. Type this matrix in MATLAB and use MATLAB to carry out the following instructions.

$$A = \begin{bmatrix} 3 & 7 & -4 & 12 \\ -5 & 9 & 10 & 2 \\ 6 & 13 & 8 & 11 \\ 15 & 5 & 4 & 1 \end{bmatrix}$$

- a. Create a 4×3 array **B** consisting of all elements in the second through fourth columns of **A**.
 - b. Create a 3×4 array **C** consisting of all elements in the second through fourth rows of **A**.
 - c. Create a 2×3 array **D** consisting of all elements in the first two rows and the last three columns of **A**.
- 7.* Compute the length and absolute value of the following vectors:
- a. **x** = [2, 4, 7]
 - b. **y** = [2, -4, 7]
 - c. **z** = [5 + 3i, -3 + 4i, 2 - 7i]

8. Given the matrix

$$A = \begin{bmatrix} 3 & 7 & -4 & 12 \\ -5 & 9 & 10 & 2 \\ 6 & 13 & 8 & 11 \\ 15 & 5 & 4 & 1 \end{bmatrix}$$

- Find the maximum and minimum values in each column.
- Find the maximum and minimum values in each row.

9. Given the matrix

$$A = \begin{bmatrix} 3 & 7 & -4 & 12 \\ -5 & 9 & 10 & 2 \\ 6 & 13 & 8 & 11 \\ 15 & 5 & 4 & 1 \end{bmatrix}$$

- Sort each column and store the result in an array **B**.
- Sort each row and store the result in an array **C**.
- Add each column and store the result in an array **D**.
- Add each row and store the result in an array **E**.

10. Consider the following arrays.

$$A = \begin{bmatrix} 1 & 4 & 2 \\ 2 & 4 & 100 \\ 7 & 9 & 7 \\ 3 & \pi & 42 \end{bmatrix} \quad B = \ln(A)$$

Write MATLAB expressions to do the following.

- Select just the second row of **B**.
- Evaluate the sum of the second row of **B**.
- Multiply the second column of **B** and the first column of **A** element by element.
- Evaluate the maximum value in the vector resulting from element-by-element multiplication of the second column of **B** with the first column of **A**.
- Use element-by-element division to divide the first row of **A** by the first three elements of the third column of **B**. Evaluate the sum of the elements of the resulting vector.

Problem#5

```

A =
[[ 3  7 -4 12]
 [-5  9 10  2]
 [ 6 13  8 11]
 [15  5  4  1]]

v = [ 7  9 13  5]

w = [-5  9 10  2]

```

=====

 Problem#6

```

A =
[[ 3  7 -4 12]
 [-5  9 10  2]
 [ 6 13  8 11]
 [15  5  4  1]]

a =
[[ 7 -4 12]
 [ 9 10  2]
 [13  8 11]
 [ 5  4  1]]

b =
[[-5  9 10  2]
 [ 6 13  8 11]
 [15  5  4  1]]

c =
[[ 7 -4 12]
 [ 9 10  2]]

```

=====

 Problem#7

```

Length = 3
Absolute Value = [2 4 7]
Magnitude of the vector = 8.306623862918075
Magnitude of the vector = 8.306623862918075

```

```

Length = 3
Absolute Value = [2 4 7]
Magnitude of the vector = 8.306623862918075
Magnitude of the vector = 8.306623862918075

```

```

Real Part = [ 5. -3.  2.]
Imaginary Part = [ 3.  4. -7.]
Length = 3
Absolute Value = [5.83095189 5.          7.28010989]

```

Problem#8

```
[[ 3  7 -4 12]
 [-5  9 10  2]
 [ 6 13  8 11]
 [15  5  4  1]]
```

```
Maximum of each column = [15 13 10 12]
Minimum of each column = [-5  5 -4  1]
```

```
Maximum of each row = [12 10 13 15]
Minimum of each row = [-4 -5  6  1]
```

=====

 Problem#9

```
[[ 3  7 -4 12]
 [-5  9 10  2]
 [ 6 13  8 11]
 [15  5  4  1]]
```

- Sort each column and store the result in an array **B**.
- Sort each row and store the result in an array **C**.
- Add each column and store the result in an array **D**.
- Add each row and store the result in an array **E**.

a

```
array([[ -5,  5, -4,  1],
       [  3,  7,  4,  2],
       [  6,  9,  8, 11],
       [15, 13, 10, 12]])
```

b

```
array([[ -4,  3,  7, 12],
       [-5,  2,  9, 10],
       [  6,  8, 11, 13],
       [  1,  4,  5, 15]])
```

c

```
array([19, 34, 18, 26])
```

d

```
array([18, 16, 38, 25])
```

Problem#10

10. Consider the following arrays.

$$A = \begin{bmatrix} 1 & 4 & 2 \\ 2 & 4 & 100 \\ 7 & 9 & 7 \\ 3 & \pi & 42 \end{bmatrix} \quad B = \ln(A)$$

Write MATLAB expressions to do the following.

- Select just the second row of **B**.
- Evaluate the sum of the second row of **B**.
- Multiply the second column of **B** and the first column of **A** element by element.
- Evaluate the maximum value in the vector resulting from element-by-element multiplication of the second column of **B** with the first column of **A**.
- Use element-by-element division to divide the first row of **A** by the first three elements of the third column of **B**. Evaluate the sum of the elements of the resulting vector.

A Matrix

```
[[ 1.      4.      2.      ]
 [ 2.      4.     100.     ]
 [ 7.      9.      7.      ]
 [ 3.      3.14159265 42.     ]]
```

B Matrix

```
array([[0.      , 1.38629436, 0.69314718],
       [0.69314718, 1.38629436, 4.60517019],
       [1.94591015, 2.19722458, 1.94591015],
       [1.09861229, 1.14472989, 3.73766962]])
```

a

```
array([0.69314718, 1.38629436, 4.60517019])
```

b

```
6.684611727667928
```

c

```
Second Column of B = [1.38629436 1.38629436 2.19722458 1.14472989]
First Column of A = [1. 2. 7. 3.]
Answer = [ 1.38629436 2.77258872 15.38057204 3.43418966]
```

d

```
15.380572041353537
```

e

```
Frist row of A = [1. 4. 2.]
First 3 elements of third column of B = [0.69314718 4.60517019 1.94591015]
Answer = [1.44269504 0.86858896 1.02779668]
```